

HAT-P-36b

R-0658

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260531 · created 2026-07-27T17:28:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461191.8059 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1575 +/- 0.0067
Transit depth (Rp/Rs)^2	2.48 +/- 0.21 [%]
Orbital Inclination (inc)	89.59 +/- 1.5
Airmass coefficient 1 (a1)	1.359 +/- 0.019
Airmass coefficient 2 (a2)	-0.141 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	None
Optimal Aperture	3.59
Optimal Annulus	6.28
Transit Duration (day)	0.10021 +/- 0.00099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1575 $\pm$ 0.0067 vs 0.126 (deviation 4.66 σ)
Duration fit vs published	0.10021 d vs 0.0925 d (deviation 7.72 σ)
Mid-time O-C	1.1 min
Depth SNR	23.3
Residual scatter	1.39 %

Light curve

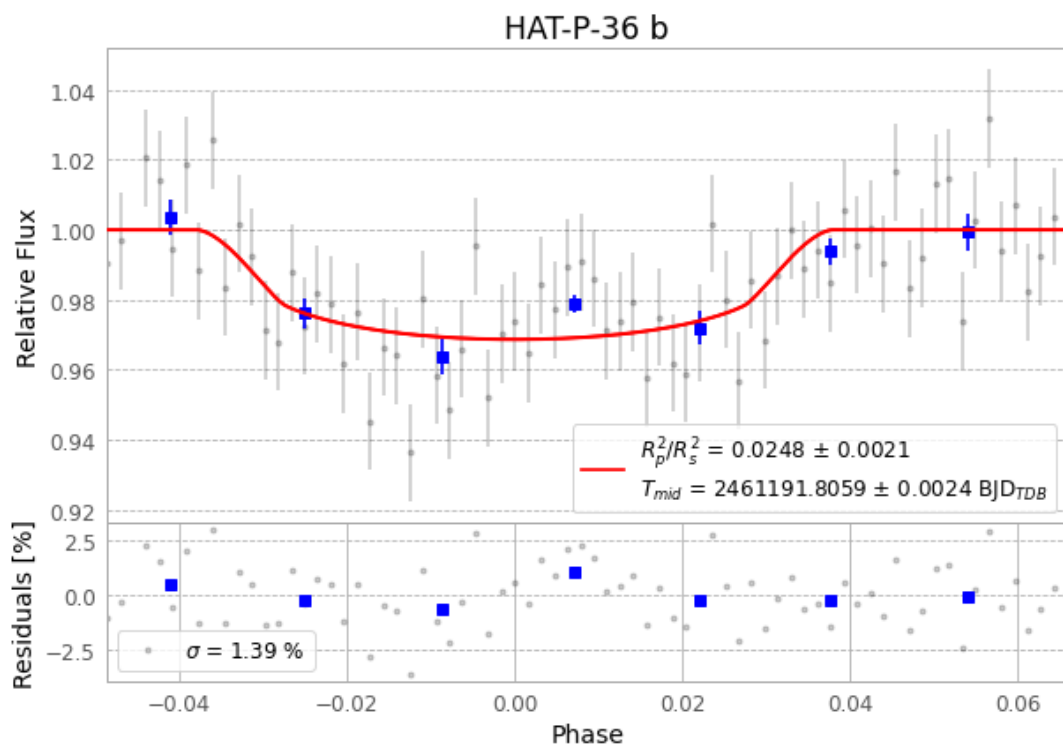


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-05-30.png (SHA-256 2c64b8bc98a00324...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 193], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 747f3f166ed79c30...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

74 FITS frames (49 MB), HATP-36260531054515.FITS → HATP-36260531092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260531.log (a931f0134a2f...), FinalLightCurve_HAT-P-36 b_2026-05-30.png (2c64b8bc98a0...), FinalLightCurve_HAT-P-36 b_2026-05-30.csv (d723156ef0d0...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:28Z.